ST FRANCIS CATHOLIC SECONDARY SCHOOL, LAGOS

The Jesuit-Run Catholic College

HORTICUTURE AND CROP PRODUCTION

JSS 1

First Term

2026/2027

JSS 1: SCHEME OF WORK FOR FIRST TERM (HORTIUCLTURE AND CROP PRODUCTION)

WEEK	TOPIC	CONTENT
1.	Introduction to crop production	A. Meaning of crops: definition of crops, and crop production B. Importance of Crop Production i. Roles of crops to man C. Classification of crops: i. Reasons for classifying crops
2.	Classification of Crops.	i. Based on lifespan ii. Based on morphology
3.	Classification of Crops	iii. Based on agronomic uses. iv. Differences between crop types
4.	Horticulture	A. Meaning and importance of horticulture B. Branches of horticulture i. Floriculture(flowers) ii. Olericulture(vegetables) iii. Pomology(fruits) iv. Arboriculture(trees) v. Viticulture(grapes) vi. Landscaping and gardening
5.	Cultural Practices	A. Meaning and Importance B. Pre-planting Operations i. Site selection: Factors to Consider when siting a Farm ii. Soil texture and Crop Types
6.	Cultural Practices	C. Land Preparation Practices: clearing, stumping, plotting, tilling, leveling, fertilizer application
7.	Simple Farm Tools	A. Meaning and Importance B. Farm tools used for pre-planting operation
8	Cultural Practices	A. Planting Operation: Meaning and Activities B. Crop Propagation: Meaning and types. C. Sexual Propagation: Meaning, Importance and Processes
9	Crop Propagation	A. Crop Propagation: Meaning and types. B. Sexual Propagation: Meaning, Importance and Processes
10	Asexual Propagation	A. Natural Asexual Propagation B. Artificial Asexual Propagation
11	REVISION	REVISION
12	EXAMINATION	EXAMINATION

WEEK1

DATE	
Topic	**Introduction to Crop Production**
Objectives / Learning Outcomes	By the end of the lesson students should be able to: i. Define crops and crop production. ii. State the importance of crop production. iii. Explain the roles of crops in human life. iv. Classify crops into different groups. v. State reasons for classifying crops.
Insight / Key Content	Every meal we eat, the clothes we wear, and many products we use daily originate from crops. Crop production is the foundation of agriculture and plays a vital role in feeding the population, supplying raw materials for industries, creating employment, and supporting national development. Understanding crops and their classification helps farmers make better production and management decisions.
Key Words / New Terms	Crops, Crop Production, Food Crops, Cash Crops, Classification, Agriculture, Raw Materials, Livestock Feed
Application	Identifying crops commonly grown in the community; grouping crops according to their uses; recognizing how crops contribute to daily life and the economy.
Summary	Crops are cultivated plants grown for food, income, raw materials, medicine, and other purposes. Crop production is important because it provides food, employment, industrial materials, and economic benefits. Crops can be classified into different groups based on their characteristics and uses.
Student Engagement **i. What I Have Learnt:** **ii. Skills Acquired:**	
My Questions	
Action:	

WEEK 1

INTRODUCTION TO CROP PRODUCTION

1. Meaning of Crops

A crop is any plant that is deliberately cultivated by human beings for food, income, raw materials, medicine, decoration, or other useful purposes.

Crops are grown on farms, gardens, plantations, and orchards to meet human needs.

Examples of Crops

- Maize
- Rice
- Yam
- Cassava
- Beans
- Tomato
- Pepper
- Cocoa
- Cotton
- Oil Palm

2. Meaning of Crop Production

Crop production is the branch of agriculture that deals with the cultivation, management, harvesting, processing, and marketing of crops for human and industrial use.

It involves all activities carried out from land preparation to harvesting and storage of crops.

Activities Involved in Crop Production

- Site selection
- Land clearing
- Tilling
- Planting
- Weeding
- Fertilizer application
- Pest and disease control
- Harvesting

- Storage

3. Importance of Crop Production

A. Provides Food

Many crops serve as food for humans. Examples:

- Rice
- Yam
- Cassava
- Maize
- Beans

B. Provides raw materials for Industries

Many industries depend on crops for production.

Examples:

Crop	Industry Product
Cotton	Textile materials
Cocoa	Chocolate
Oil Palm	Soap and cooking oil
Sugarcane	Sugar
Rubber	Tyre

C. Creates Employment

Crop production provides jobs for:

- Farmers
- Traders
- Transporters
- Processors
- Agricultural extension workers

D. Generates Income

Farmers earn money from selling crops and crop products.

E. Provides Foreign Exchange

Export crops such as cocoa, rubber, sesame, and cashew generate foreign exchange earning for the country.

F. Provides Livestock Feed

Some crops are used for feeding farm animals.

Examples:

- Maize
- Sorghum
- Soybean

4. Roles of Crops to Man

Crops play several important roles in human life.

A. Source of Food

Most of the food consumed by humans come from crops.

B. Source of Shelter

Some crops provide building materials.

Examples:

- Bamboo
- Raffia palm

C. Source of Clothing

Fibres obtained from crops are used for making clothes.

Example:

- Cotton

D. Source of Medicine

Many medicinal products are obtained from plants.

Examples:

- Aloe vera
- Neem
- Ginger

E. Source of Fuel

Some crops provide fuel and energy.

Examples:

- Firewood
- Charcoal
- Biofuel crops

F. Environmental Protection

Plants help to:

- Reduce soil erosion
- Improve air quality
- Provide shade
- Beautify the environment

5. Classification of Crops

Crop classification is the grouping of crops according to their similarities and characteristics.

This helps farmers, researchers, and students study crops more effectively.

6. Reasons for Classifying Crops

A. Easy Identification: Classification makes it easier to identify crops.

B. Better Study of Crops: It helps students and farmers understand crop characteristics**.**

C. Proper Farm Management: Different crops require different management practices.

D. Easy Communication: Farmers and scientists can easily discuss crops using standard groups.

E. Agricultural Planning: Classification helps in planning crop production and utilization.

Key Points to Remember

- A crop is a useful plant cultivated by humans for specific purposes.
- Crop production involves all activities from planting to harvesting of crops.
- Crops provide food, income, employment, medicine, clothing, and raw materials.
- Crops are classified into groups based on their characteristics and uses

WEEK 2

DATE	
Topic	**Classification of Crops (Based on Lifespan and Morphology)**
Objectives / Learning Outcomes	By the end of the lesson, students should be able to: i. Define crop classification. ii. Explain classification of crops based on lifespan. iii. Differentiate between annual, biennial, and perennial crops. iv. Explain classification of crops based on morphology. v. Identify examples of crops under each classification.
Insight / Key Content	Not all crops grow and live for the same length of time, nor do they have the same physical structure. Some crops complete their life cycle within a year, while others live for many years.
Key Words / New Terms	Crop Classification, Lifespan, Annual Crops, Biennial Crops, Perennial Crops, Morphology, Cereals, Legumes, Root Crops
Application	Identifying crops in the school farm and their locality according to their lifespan and physical characteristics; grouping crops correctly for farming and academic purposes.
Summary	Crops can be classified according to their lifespan and morphology. Based on lifespan, crops are grouped into annuals, biennials, and perennials. Based on morphology, crops are grouped according to their physical characteristics such as cereals, legumes, root and tuber crops, fruits, and vegetables.
Student Engagement **i. What I Have Learnt:** **ii. Skills Acquired:**	
My Question(s)	
Action:	

WEEK 2

CLASSIFICATION OF CROPS

Meaning of Crop Classification

Crop classification is the grouping of crops according to their similarities and characteristics.

Classification helps farmers, students, and researchers to study crops more effectively and apply suitable management practices.

A. Classification of Crops Based on Lifespan

Lifespan refers to the length of time a crop takes to complete its life cycle from germination to death.

Based on lifespan, crops are classified into:

1. Annual Crops

Annual crops complete their life cycle within one year or one growing season.

They germinate, grow, flower, produce seeds, and die within a short period.

Examples

- Maize
- Rice
- Tomato
- Okra
- Groundnut
- Cowpea

Characteristics

- Short lifespan
- Mature quickly
- Replanted every year

2. Biennial Crops

Biennial crops require two years to complete their life cycle.

During the first year, they produce leaves and roots, while flowering and seed production occur in the second year.

Examples

- Carrot
- Cabbage
- Onion
- Beetroot

Characteristics

- Complete life cycle in two years
- Store food during the first year
- Produce flowers and seeds during the second year

3. Perennial Crops

Perennial crops live for more than two years and continue producing for many years.

Examples

- Cocoa
- Coffee
- Oil Palm
- Mango
- Orange
- Rubber

Characteristics

- Long lifespan
- Produce yields repeatedly
- Require long-term management

Summary Table

Crop Type	Lifespan	Examples
Annual	Less than one year	Maize, Rice, Tomato
Biennial	Two years	Onion, Carrot, Cabbage
Perennial	More than two years	Cocoa, Mango, Oil Palm

B. Classification of Crops Based on Morphology

Morphology refers to the external structure and form of plants.

Based on morphology, crops are classified into:

1. Monocotyledonous Crops (Monocots)
2. Dicotyledonous Crops (Dicots)

1. Monocotyledonous Crops (Monocots)

These are crops whose seeds contain only one cotyledon (seed leaf).

Characteristics of Monocots

- One cotyledon
- Fibrous root system
- Parallel leaf venation
- Floral parts or their flower petals occur in multiples of three
- Stem usually has scattered vascular bundles

Examples

- Maize
- Rice
- Millet
- Sorghum
- Wheat
- Coconut

2. Dicotyledonous Crops (Dicots)

These are crops whose seeds contain two cotyledons (seed leaves).

Characteristics of Dicots

- Two cotyledons
- Tap root system
- Reticulate (net-like) leaf venation
- Floral parts occur in multiples of four or five
- Vascular bundles arranged in a ring form

Examples

- Cowpea
- Groundnut
- Soybean
- Mango
- Orange
- Cocoa
- Tomato

Differences between Monocots and Dicots

Monocots	Dicots
One cotyledon	Two cotyledons
Fibrous roots	Tap roots
Parallel venation	Net-like venation
Floral parts in threes	Floral parts in fours or fives
Maize, Rice	Cowpea, Groundnut

Importance of Crop Classification

Crop classification helps to:

1. Identify crops easily.
2. Study crop characteristics.
3. Improve crop management.
4. Facilitate agricultural research.
5. Enhance communication among farmers and scientists.

Key Points to Remember

- Crops can be classified according to lifespan and morphology.
- Based on lifespan, crops are annual, biennial, and perennial.
- Based on morphology, crops are monocotyledons and dicotyledons.
- Monocots have one cotyledon and fibrous roots.
- Dicots have two cotyledons and tap roots.

WEEK3

DATE	
Topic	**Classification of Crops Based on Agronomic Uses and Differences Between Crop Types**
Objectives / Learning Outcomes	By the end of the lesson, students should be able to: i. Explain classification of crops based on agronomic uses. ii. Identify different crop groups according to their uses. iii. Give examples of crops under each agronomic class. iv. Differentiate between various crop types. v. State the importance of agronomic classification in agriculture.
Insight	Although crops are all cultivated plants, they serve different purposes in agriculture and human life. Some provide food, others supply raw materials for industries, medicine, fibre, beverages, livestock feed, or oils.
Key Words / New Terms	Agronomic Uses, Cereals, Legumes, Root Crops, Tuber Crops, Oil Crops, Fibre Crops, Beverage Crops, Forage Crops, Medicinal Crops
Application	Grouping crops according to their uses; identifying crops grown in the school farm, their locality and economic importance; selecting suitable crops for specific agricultural purposes.
Summary	Crops can be classified according to their agronomic uses into cereals, legumes, root and tuber crops, oil crops, fibre crops, beverage crops, forage crops, and medicinal crops. Each group serves a unique purpose and contributes to food production, industrial development, and economic growth.
Student Engagement **i. What I Have Learnt:** **ii. Skills Acquired**	
My Question(s)	
Action:	

WEEK 3

CLASSIFICATION OF CROPS BASED ON AGRONOMIC USES

Meaning of Agronomic Classification

Agronomic classification is the grouping of crops according to their economic importance and agricultural uses.

This classification helps farmers understand the purpose of different crops and how they contribute to food production, industry, and national development.

Classes of Crops Based on Agronomic Uses

1. Cereal Crops

Cereals are crops cultivated mainly for their grains.

They are rich in carbohydrates and serve as major sources of food for humans and livestock.

Examples

- Maize
- Rice
- Millet
- Sorghum
- Wheat

Importance

- Source of energy
- Human food
- Livestock feed
- Industrial raw materials

2. Legume Crops

Legumes are crops that produce seeds in pods.

They are rich in protein and improve soil fertility through nitrogen fixation.

Examples

- Cowpea
- Groundnut
- Soybean
- Pigeon pea
- Bambara nut

- Beans

Importance

- Source of protein
- Livestock feed
- Soil improvement

3. Root and Tuber Crops

These are crops grown mainly for their enlarged underground roots or underground stems.

Examples

Root Tuber Crops

- Cassava
- Sweet potato

Stem Tuber Crops

- Yam
- Irish potato

Importance

- Source of carbohydrates
- Food for humans
- Industrial processing

4. Oil Crops

These are crops cultivated mainly for the extraction of oil.

Examples

- Oil palm
- Groundnut
- Coconut
- Soybean
- Sesame

Importance

- Production of cooking oil
- Soap manufacturing
- Industrial uses

5. Fibre Crops

These crops provide fibres used in making textiles, ropes, bags, and other materials.

Examples

- Cotton
- Jute
- Kenaf
- Sisal

Importance

- Textile production
- Rope making
- Industrial raw materials

6. Beverage Crops

These crops are used in the production of drinks and stimulants.

Examples

- Cocoa
- Coffee
- Tea

- Kola

Importance

- Beverage production
- Export earnings
- Industrial processing

7. Forage Crops

These are crops grown mainly for feeding livestock.

Examples

- Elephant grass
- Guinea grass
- Centrosema
- Stylo

Importance

- Livestock nutrition
- Pasture development

8. Medicinal Crops

These crops are valued for their healing and medicinal properties.

Examples

- Aloe vera
- Neem
- Ginger
- Garlic
- Turmeric

Importance

- Production of medicines
- Traditional healthcare

9. Spices and Condiment Crops These are crops used to improve the flavour, taste, colour, and aroma of food. Examples

* Pepper
* Ginger
* Garlic
* Onion
* Cloves

Importance

* Improve food taste and aroma
* Used in food processing
* Some have medicinal value

10. Aromatic Crops: These are crops grown mainly for their pleasant scent or essential oils.
Examples

Lemongrass
Mint
Basil
Lavender
Eucalyptus

Importance
* Production of perfumes
* Production of essential oils
* Used in cosmetics and medicine
* Used in food and beverage industries

Differences Between Crop Types

Different crop types can be distinguished by their uses and economic importance.

Crop Type	Main Use	Examples
Cereals	Source of carbohydrates	Maize, Rice
Legumes	Source of protein	Cowpea, Soybean
Root and Tuber Crops	Energy food	Yam, Cassava
Oil Crops	Oil production	Oil Palm, Coconut
Fibre Crops	Textile materials	Cotton, Kenaf
Beverage Crops	Drink production	Cocoa, Coffee
Forage Crops	Animal feed	Guinea Grass, Stylo
Medicinal Crops	Medicine production	Neem, Aloe Vera

Importance of Agronomic Classification

Agronomic classification helps to:

1. Identify the economic value of crops.
2. Select crops for specific purposes.
3. Improve farm planning.
4. Promote efficient land use.
5. Facilitate agricultural research and extension services.

Key Points to Remember

- Agronomic classification groups crops according to their uses.
- Major agronomic groups include cereals, legumes, root and tuber crops, oil crops, fibre crops, beverage crops, forage crops, and medicinal crops.
- Different crop groups serve different purposes in agriculture and industry.
- Agronomic classification assists farmers in proper crop selection and management.

WEEK 4

DATE	
Topic	**Horticulture**
Objectives / Learning Outcomes	By the end of the lesson, students should be able to: i. Define horticulture. ii. State the importance of horticulture. iii. Identify the major branches of horticulture. iv. Explain the branches of horticulture. v. Give examples of crops under each branch.
Insight	Horticulture is one of the most visible branches of agriculture because it deals with the fruits, vegetables, flowers, and ornamental plants that people interact with every day.
Key Words / New Terms	Horticulture, Floriculture, Olericulture, Pomology, Arboriculture, Viticulture, Landscaping, Gardening, Ornamental Plants
Application	Identifying horticultural crops in SFCSS, homes and community; establishing home gardens; beautifying the environment with flowers and ornamental plants; growing fruits and vegetables for food and income.
Summary	Horticulture is the branch of agriculture concerned with the cultivation of fruits, vegetables, flowers, and ornamental plants. Its major branches include floriculture, olericulture, pomology, arboriculture, viticulture, landscaping, and gardening. Horticulture provides food, income, employment, and environmental beautification.
Student Engagement **i. What I Have Learnt:** **ii. Skills Acquired:**	
My Question(s)	
Action:	

WEEK 4

HORTICULTURE

Meaning of Horticulture

Horticulture is a branch of agriculture that deals with the cultivation, production, improvement, processing, and marketing of fruits, vegetables, flowers, ornamental plants, and trees.

The word "horticulture" is derived from two Latin words:

- *Hortus* meaning garden
- *Cultura* meaning cultivation

Therefore, horticulture means the cultivation of garden crops.

Importance of Horticulture

Horticulture contributes greatly to human welfare and economic development.

1. Source of Food

Horticultural crops provide fruits and vegetables rich in vitamins, minerals, and other nutrients needed for good health.

Examples include:

- Orange
- Mango
- Tomato
- Cabbage
- Carrot

2. Source of Income

Farmers earn income from the sale of fruits, vegetables, flowers, and ornamental plants.

3. Employment Generation

Horticulture provides employment opportunities for:

- Farmers
- Gardeners
- Florists
- Landscapers
- Fruit sellers
- Nursery operators

4. Environmental Beautification

Flowers, lawns, and ornamental plants beautify homes, schools, offices, parks, and roadsides.

5. Raw Materials for Industries

Many horticultural crops are processed into industrial products**.**

Examples:

- **Tomato paste**
- **Fruit juice**
- **Jam**
- **Wine**

6. Improvement of Human Health

Regular consumption of fruits and vegetables helps prevent nutritional deficiencies and promotes healthy living**.**

Branches of Horticulture

Horticulture is divided into several branches based on the crops cultivated.

1. Floriculture

Floriculture is the cultivation and management of flowers and ornamental plants.

Examples

- **Rose**
- **Hibiscus**
- **Sunflower**
- **Bougainvillea**
- **Marigold**

Importance

- Beautification
- Decoration
- Landscaping
- Income generation

2. Olericulture

Olericulture is the cultivation and production of vegetable crops.

Examples

- Tomato
- Pepper
- Cabbage
- Lettuce
- Okra
- Spinach

Importance

- Food supply
- Nutrition
- Income generation

3. Pomology

Pomology is the cultivation and study of fruit crops**.**

Examples

- **Mango**
- **Orange**

- **Pawpaw**
- **Guava**
- **Pineapple**
- **Avocado**

Importance

- Source of vitamins
- Food and nutrition
- Export earnings

4. Arboriculture

Arboriculture is the cultivation, management, and study of trees and shrubs.

Examples

- Neem
- Mahogany
- Casuarina
- Ornamental trees

Importance

- Shade provision
- Environmental protection
- Beautification

5. Viticulture

Viticulture is the cultivation and management of grape vines.

Examples

- Table grapes
- Wine grapes

Importance

- Fruit production
- Juice production
- Wine industry

6. Landscaping and Gardening

Landscaping

Landscaping is the art and science of arranging plants and other materials to create attractive outdoor environments.

Importance of Landscaping

- Beautifies the environment
- Reduces soil erosion
- Improves air quality
- Creates recreational spaces

Gardening

Gardening is the cultivation of plants around homes, schools, and other locations for food, beauty, or recreation.

Types of Gardens

- Home garden
- School garden
- Vegetable garden
- Flower garden

Summary Table of Branches of Horticulture

Branch	Meaning	Examples
Floriculture	Cultivation of flowers	Rose, Hibiscus
Olericulture	Cultivation of vegetables	Tomato, Cabbage
Pomology	Cultivation of fruits	Mango, Orange
Arboriculture	Cultivation of trees	Neem, Mahogany
Viticulture	Cultivation of grapes	Grapes
Landscaping	Beautification using plants	Lawns, Flower beds
Gardening	Growing plants around homes and schools	Vegetables, Flowers

Key Points to Remember

1. **Horticulture is the cultivation of fruits, vegetables, flowers, and ornamental plants.**
2. **Horticulture provides food, income, employment, and environmental beauty.**
3. **Floriculture deals with flowers.**
4. **Olericulture deals with vegetables.**
5. **Pomology deals with fruits.**
6. **Arboriculture deals with trees.**
7. **Viticulture deals with grapes.**
8. **Landscaping and gardening improve the beauty and usefulness of the environment.**

WEEK 5

DATE:	
Topic	**Cultural Practices**
Objectives / Learning Outcomes	**By the end of the lesson, students should be able to:** i. Define cultural practices. ii. Explain the importance of cultural practices in crop production. iii. Define site selection. iv. State factors to consider when selecting a farm site. v. Explain soil texture and its relationship with crop production. vi. Match suitable crops with different soil types.
Insight	Successful crop production begins long before planting. Choosing the right farm site and understanding the nature of the soil are important decisions that determine whether crops will thrive or fail.
Key Words / New Terms	Cultural Practices, Site Selection, Soil Texture, Sandy Soil, Clay Soil, Loamy Soil, Crop Yield, Drainage, Fertility
Application	Selecting suitable locations for SFCSS farm, farms; identifying soil types in the school farm and environment; recommending suitable crops for different soil conditions; improving crop productivity through proper planning.
Summary	Cultural practices are activities carried out to ensure successful crop production. Proper site selection and knowledge of soil texture help farmers choose suitable crops, improve growth conditions, and increase crop yield.
Student Engagement **i. What I Have Learnt:** **ii. Skills Acquired:**	

My Question(s)	
Action:	

WEEK 5

CULTURAL PRACTICE

Meaning of Cultural Practices

Cultural practices are the various activities carried out by farmers before, during, and after planting to ensure healthy crop growth and good yield.

These practices help crops obtain adequate nutrients, water, sunlight, and protection from weeds, pests, and diseases.

Examples of cultural practices include:

Cultural Practices	Activities
Pre-planting	● Site selection ● Land preparation ● Irrigation
Planting	● Planting ● Irrigation

Post-Planting	● Weeding ● Mulching ● Fertilizer application ● Irrigation ● Harvesting

Importance of Cultural Practices

Cultural practices are important because they:

1. Improve crop growth and development.
2. Increase crop yield.
3. Reduce weed competition.
4. Help control pests and diseases.
5. Improve soil fertility.
6. Ensure efficient use of land and resources.
7. Promote healthy and quality crops**.**

Site Selection

Site selection is the process of choosing a suitable piece of land for crop production.

A good farm site provides favourable conditions for crop growth and high productivity.

Proper site selection is one of the most important decisions a farmer makes before planting**.**

Factors to Consider When Siting a Farm

1. Soil Fertility

The soil should contain adequate nutrients required for plant growth.

Fertile soils support healthy crops and good yields.

2. Soil Texture

The soil should have a suitable texture that supports root growth, water retention, and aeration.

Loamy soil is generally considered the best for most crops.

3. Topography

Topography refers to the shape or slope of the land.

Gentle slopes are preferred because they reduce erosion and facilitate farm operations.

4. Availability of Water

A reliable source of water is necessary for crop growth, especially during dry periods.

Sources may include:

- Rivers
- Streams
- Wells
- Boreholes
- Rainfall

5. Climate

Temperature, rainfall, humidity, and sunshine should be suitable for the crops to be cultivated.

Different crops require different climatic conditions**.**

6. Accessibility

The farm should be easily accessible for transportation of inputs and harvested produce.

Good roads reduce transportation costs**.**

7. Security

The farm should be protected from theft, destruction by animals, and other forms of damage.

8. Availability of Labour

Adequate labour should be available for farm operations**.**

Soil Texture

Soil texture refers to the relative proportion of sand, silt, and clay particles in a soil.

Soil texture affects:

- Water retention
- Drainage
- Aeration
- Root penetration
- Nutrient availability

Types of Soil Based on Texture

1. Sandy Soil

Sandy soil contains a high proportion of sand particles.

Characteristics

- Large particles
- Drains water quickly
- Well aerated
- Low water-holding capacity
- Low nutrient retention

Suitable Crops

- Groundnut
- Watermelon
- Coconut
- Cashew

2. Clay Soil

Clay soil contains a high proportion of clay particles.

Characteristics

- Very small particles
- Holds water for a long time
- Poor drainage
- High nutrient retention
- Becomes sticky when wet

Suitable Crops

- Rice
- Sugarcane

3. Loamy Soil

Loamy soil contains a balanced mixture of sand, silt, and clay.

Characteristics

- Good drainage
- Good water retention
- Rich in nutrients
- Well aerated
- Easy to cultivate

Suitable Crops

- Maize
- Tomato
- Yam
- Cassava
- Vegetables
- Fruit crops

Loamy soil is regarded as the best soil for crop production**.**

Relationship Between Soil Texture and Crop Types

Different crops grow best in different soil textures.

Soil Type	Suitable Crops
Sandy Soil	**Groundnut, Watermelon, Coconut**
Clay Soil	**Rice, Sugarcane**
Loamy Soil	**Maize, Yam, Cassava, Tomato, Vegetables**

Farmers should select crops that are suitable for the texture of the soil available on the farm.

Key Points to Remember

1. Cultural practices are activities carried out to ensure successful crop production.
2. Site selection is the choice of suitable land for farming.
3. Important factors in farm siting include soil fertility, water availability, climate, accessibility, and security.
4. Soil texture refers to the proportion of sand, silt, and clay in the soil.
5. Sandy soil, clay soil, and loamy soil have different characteristics.
6. Different crops perform best under different soil conditions.
7. Loamy soil is generally the most suitable soil for crop production.

WEEK 6

Topic	**Cultural Practices: Land Preparation Practices**
Objectives / Learning Outcomes	By the end of the lesson, students should be able to: i. Define land preparation. ii. Explain the importance of land preparation in crop production. iii. Identify the various land preparation practices. iv. Describe clearing, stumping, plotting, tilling, levelling, and fertilizer application. v. State the benefits of proper land preparation.
Insight	A building cannot stand on a poor foundation, and crops cannot thrive on poorly prepared land. The success of a farming season often depends on how well the land is prepared before planting.
Key Words / New Terms	Land Preparation, Clearing, Stumping, Plotting, Tilling, Levelling, Fertilizer Application, Seedbed, Soil Fertility
Application	Participating in school farm activities; identifying stages of land preparation; preparing small garden plots for crop cultivation; applying appropriate land preparation methods for different crops.
Summary	Land preparation involves all activities carried out to make the land suitable for planting. Major land preparation practices include clearing, stumping, plotting, tilling, levelling, and fertilizer application. Proper land preparation promotes healthy crop growth and improves yield.
Student Engagement **i. What I Have Learnt:** **ii. Skills Acquired:**	

My Question(s)	
Action:	

WEEK 6

CULTURAL PRACTICES: LAND PREPARATION

Meaning of Land Preparation

Land preparation refers to all the activities carried out on farmland before planting in order to make the soil suitable for crop growth.

It involves removing obstacles, improving soil conditions, and creating a favourable environment for seed germination and plant development.

Importance of Land Preparation

Land preparation is important because it:

1. Creates a suitable seedbed for planting.
2. Promotes easy germination of seeds.
3. Improves soil aeration.
4. Enhances water infiltration and retention.
5. Controls weeds and unwanted vegetation.
6. Facilitates farm operations.

7. Encourages proper root development.
8. Increases crop yield.

Land Preparation Practices

1. Clearing

Clearing is the removal of grasses, weeds, shrubs, trees, and other unwanted vegetation from the farm site.

Methods of Clearing

- Manual clearing using cutlass and hoe
- Mechanical clearing using bulldozers and tractors
- Chemical clearing using herbicides

Importance of Clearing

- Reduces competition for nutrients
- Makes farm operations easier
- Exposes the land for cultivation

2. Stumping

Stumping is the removal of tree stumps, roots, and other underground obstacles from the farmland after clearing.

Importance of Stumping

- Prevents obstruction during cultivation
- Facilitates mechanized farming
- Promotes root penetration
- Makes tillage operations easier

3. Plotting

Plotting is the division of farmland into smaller sections called plots for easy management.

Importance of Plotting

- Ensures proper farm organization
- Makes crop management easier
- Facilitates experimental and demonstration activities
- Enhances proper spacing of crops

4. Tilling

Tilling is the process of loosening and turning the soil before planting.

It may be carried out manually using hoes or mechanically using ploughs and harrows**.**

Types of Tillage

Primary Tillage

The first breaking and turning of the soil.

Examples:

- Ploughing
- Digging

Secondary Tillage

Further breaking and smoothing of the soil after primary tillage. **Examples:**

- Harrowing
- Ridging

Importance of Tilling

- Improves soil aeration
- Facilitates root growth
- Enhances water penetration
- Controls weeds
- Mixes organic matter with soil

5. Levelling

Levelling is the process of making the land surface even after tillage.

Importance of Levelling

- Prevents waterlogging
- Promotes uniform water distribution
- Facilitates planting operations
- Reduces soil erosion

6. Fertilizer Application

Fertilizer application is the addition of nutrients to the soil to improve its fertility and support plant growth.

Types of Fertilizers

Organic Fertilizers

- Farmyard manure
- Compost manure
- Green manure

Inorganic Fertilizers

- NPK fertilizer
- Urea
- Ammonium sulphate

Importance of Fertilizer Application

- Replaces lost nutrients
- Improves soil fertility
- Promotes healthy plant growth
- Increases crop yield

Sequence of Land Preparation Operations

The common sequence of land preparation is:

1. Clearing
2. Stumping
3. Plotting
4. Tilling
5. Levelling

6. Fertilizer Application
7. Planting

Proper execution of these operations helps ensure successful crop production.

Key Points to Remember

1. Land preparation consists of activities carried out before planting.
2. Clearing removes unwanted vegetation.
3. Stumping removes tree stumps and roots.
4. Plotting divides farmland into manageable units.
5. Tilling loosens and turns the soil.
6. Levelling creates an even land surface.
7. Fertilizer application improves soil fertility.
8. Proper land preparation enhances crop growth and yield.

WEEK 7

DATE	
Topic	**Simple Farm Tools**
Objectives / Learning Outcomes	By the end of the lesson, students should be able to: i. Define simple farm tools. ii. State the importance of simple farm tools in crop production. iii. Identify farm tools used for pre-planting operations. iv. State the functions of common farm tools. v. Explain the proper care and maintenance of farm tools.
Insight	Before tractors and modern machines became common, farmers depended on simple farm tools to cultivate the land and produce food. Even today, these tools remain indispensable in school gardens, home gardens, and small-scale farms. Understanding their uses and maintenance helps farmers work efficiently, safely, and economically.
Key Words / New Terms	Farm Tools, Cutlass, Hoe, Rake, Spade, Shovel, Hand Fork, Wheelbarrow, Maintenance
Application	Identifying farm tools used at home, SFCSS farm, and community gardens; selecting appropriate tools for specific farm operations; practicing proper handling and maintenance of farm tools.
Summary	Simple farm tools are hand-operated implements used to carry out various farm activities. Common tools used for pre-planting operations include the cutlass, hoe, rake, spade, shovel, and wheelbarrow. Proper use and maintenance of these tools improve efficiency and prolong their lifespan.

Student Engagement **i. What I Have Learnt:** **ii. Skills Acquired:**	
My Question(s)	
Action:	

WEEK 7

SIMPLE FARM TOOLS

Meaning of Simple Farm Tools

Simple farm tools are hand-operated tools used by farmers to carry out various agricultural activities.

They are commonly used on small farms, school farms, home gardens, and horticultural gardens.

These tools make farm work easier, faster, and more effective**.**

Importance of Simple Farm Tools

Simple farm tools are important because they:

1. Make farm work easier.
2. Reduce human effort.
3. Save time during farm operations.
4. Improve efficiency in farming.
5. Help farmers cultivate small areas of land effectively.

6. Increase crop production.
7. Reduce the cost of farming**.**

Farm Tools Used for Pre-Planting Operations

Pre-planting operations are activities carried out before planting crops**.**

The common tools used include:

1. Cutlass

A cutlass is a long metal blade with a handle used for cutting grasses, shrubs, and small trees.

Uses of Cutlass

- Land clearing
- Weeding
- Harvesting some crops
- Cutting stakes

2. Hoe

A hoe consists of a metal blade attached to a wooden handle.

Uses of Hoe

- Tilling the soil
- Making ridges
- Weeding
- Covering planted seeds
- Mixing manure with soil

3. Rake

A rake is a tool with several teeth attached to a long handle.

Uses of Rake

- Gathering weeds
- Collecting leaves and debris
- Levelling the soil surface
- Breaking soil clods

4. Spade

A spade has a flat metal blade attached to a handle.

Uses of Spade

- Digging soil
- Turning soil
- Making planting holes
- Removing soil from one place to another

5. Shovel

A shovel has a broad, curved blade.

Uses of Shovel

- Scooping soil
- Loading manure
- Moving sand and compost
- Filling holes

6. Hand Fork

A hand fork is a small tool with strong metal prongs.

Uses of Hand Fork

- Loosening soil
- Mixing manure into soil
- Removing weeds
- Working around garden plants

7. Wheelbarrow

A wheelbarrow is a small vehicle with one wheel used for transporting materials.

Uses of Wheelbarrow

- Carrying seedlings
- Transporting manure
- Moving harvested produce

- Carrying farm tools

Care and Maintenance of Farm Tools

Proper care and maintenance increase the lifespan and efficiency of farm tools**.**

Maintenance Practices

1. Clean tools after use.
2. Dry tools before storage.
3. Oil metal parts to prevent rusting.
4. Sharpen cutting edges regularly.
5. Tighten loose nuts and handles.
6. Store tools in a dry tool shed.
7. Repair damaged tools promptly.
8. Use each tool for its correct purpose.

Safety Precautions When Using Farm Tools

1. Handle tools carefully.
2. Do not play with farm tools.
3. Wear protective clothing when necessary.
4. Carry sharp tools properly.
5. Store tools safely after use.
6. Keep tools away from children when not in use.

Summary Table of Common Farm Tools

Farm Tool	Major Use
Cutlass	**Clearing and cutting vegetation**
Hoe	**Tilling, ridging and weeding**
Rake	**Gathering debris and levelling soil**
Spade	**Digging and turning soil**
Shovel	**Scooping and moving materials**
Hand Fork	**Loosening soil and removing weeds**
Wheelbarrow	**Transportation of materials**

Key Points to Remember

1. **Simple farm tools are hand-operated farm implements.**
2. **They are used mainly in small-scale farming and gardening.**
3. **Common pre-planting tools include cutlass, hoe, rake, spade, shovel, hand fork, and wheelbarrow.**
4. **Each tool performs specific farm operations.**
5. **Proper maintenance prevents damage and prolongs the lifespan of tools.**
6. **Safety precautions must always be observed when using farm tools.**

WEEK 8

Topic	**Planting Operations: Meaning and Activities**
Objectives / Learning Outcomes	By the end of the lesson, students should be able to: i. Define planting operation. ii. Explain the importance of planting operations in crop production. iii. Identify the major activities involved in planting operations. iv. Describe the procedures for proper planting. v. State factors that influence successful planting.
Insight	A healthy harvest begins with proper planting. Even when good seeds, fertile soil, and favourable weather are available, poor planting practices can lead to low yields. Farmers therefore pay close attention to planting operations to ensure good germination, healthy crop establishment, and maximum productivity.
Key Words / New Terms	Planting Operation, Germination, Seed Viability, Plant Spacing, Planting Depth, Seed Rate, Transplanting, Crop Establishment
Application	Planting seeds in school gardens; determining correct spacing for crops; preparing seedlings for transplanting; observing proper planting methods in home gardens and farms.
Summary	Planting operations are activities carried out to establish crops in the field. They include seed selection, viability testing, determination of planting time, spacing, sowing, thinning, and transplanting. Proper planting operations promote good germination, healthy growth, and high crop yield.

Student Engagement **i. What I Have Learnt:** **ii. Skills Acquired:**	
My Question(s):	
Action	

WEEK 8

PLANTING OPERATIONS

Meaning of Planting Operations

Planting operations are the activities carried out to establish crops successfully on the farm after land preparation.

Planting is one of the most important stages in crop production because the success of a crop largely depends on how well it is planted.

Importance of Planting Operations

Planting operations are important because they:

1. Ensure proper crop establishment.

2. Promote good germination of seeds.

3. Encourage healthy crop growth.

4. Improve crop yield.

5. Reduce wastage of seeds.

6. Facilitate proper spacing of crops.

7. Ensure efficient use of farmland.

Activities Involved in Planting Operations

1. Seed Selection

Seed selection involves choosing healthy, viable, and disease-free seeds for planting**.**

Characteristics of Good Seeds

- Mature and well-developed

- Free from pests and diseases

- High germination capacity

- True to type

- Free from damage

Importance of Seed Selection

- Ensures healthy seedlings

- Improves crop yield

- Reduces crop failure

2. Seed Viability Test

A viability test is carried out to determine whether seeds are capable of germinating.

Common Method

The water test:

- Healthy seeds usually sink in water.

- Bad or damaged seeds often float**.**

Importance

- Helps farmers select good seeds.

- Reduces wastage during planting**.**

3. Determination of Planting Time/Date

Planting should be done at the appropriate time to ensure good crop performance.

Factors Affecting Planting Time

- Rainfall pattern

- Crop type

- Soil moisture

- Climate

Importance

- Ensures good germination.

- Reduces seed loss.

- Promotes healthy growth.

4. Planting Methods

Planting methods refer to the ways seeds or planting materials are introduced into the soil**.**

A. Broadcasting

Seeds are scattered randomly on the soil surface.

Advantages

- Fast and easy.

- Suitable for small seeds**.**

Disadvantages

- Uneven distribution.

- Difficult weed control**.**

B. Drilling

Seeds are planted in continuous rows.

Advantages

- Proper spacing.

- Easier management.

C. Dibbling

Seeds are planted individually into holes at specified spacing and depth.

Advantages

- Accurate seed placement.

- Reduced seed wastage.

D. Transplanting

Young seedlings raised in a nursery are transferred to the main field.

Examples

- **Tomato**

- **Pepper**

- **Cabbage**

- **Onion**

Advantages

- Better crop establishment.

- Stronger seedlings**.**

5. Spacing

Spacing refers to the distance maintained between plants and rows.

Importance of Proper Spacing

- Reduces competition for nutrients.

- Promotes good aeration.

- Facilitates farm operations.

- Improves crop yield**.**

6. Planting Depth

Planting depth is the distance from the soil surface to where the seed is placed**.**

Importance

- Promotes germination.

- Prevents seed drying.

- Protects seeds from birds and pests**.**

7. Thinning

Thinning is the removal of excess seedlings after germination.

Importance

- Reduces overcrowding.

- Provides adequate space for growth.

- Improves crop development**.**

8. Supplying

Supplying is the replacement of dead or missing seedlings after planting**.**

Importance

- Maintains the desired plant population.

- Ensures uniform crop growth**.**

Factors Affecting Successful Planting

1. Quality of seeds.

2. Soil fertility.

3. Availability of moisture.

4. Correct planting depth**.**

5. Proper spacing.

6. Suitable planting time.

7. Effective weed control**.**

Key Points to Remember

1. Planting operations are activities involved in establishing crops in the field.

2. Good seed selection is essential for successful crop production.

3. Common planting methods include broadcasting, drilling, dibbling, and transplanting.

4. Proper spacing and planting depth promote healthy crop growth.

5. Thinning removes excess seedlings while supplying replaces missing plants.

6. Successful planting leads to better crop yield and productivity**.**

Revision Questions

1. Define planting operations.

2. State five importance of planting operations.

3. Mention four characteristics of good seeds.

4. What is a seed viability test?

5. Explain broadcasting and drilling methods of planting.

6. What is transplanting?

7. State four importance of proper spacing.

8. Define thinning and supplying.

9. Mention five factors affecting successful planting.

10. Why is proper planting important in crop production?

WEEK 9

Topic	**Crop Propagation: Meaning and Types; Sexual Propagation**
Objectives / Learning Outcomes	By the end of the lesson, students should be able to: i. Define crop propagation. ii. State the types of crop propagation. iii. Explain sexual propagation. iv. State the importance of sexual propagation. v. Describe the processes involved in sexual propagation.
Insight	Every crop grown on a farm begins with reproduction. Farmers use propagation techniques to multiply plants and ensure continuous food production. Sexual propagation, which involves the use of seeds, remains the most common method of raising many agricultural crops.
Key Words / New Terms	Crop Propagation, Sexual Propagation, Asexual Propagation, Seed, Pollination, Fertilization, Germination, Seedling
Application	Raising crops from seeds in SFCSS garden; selecting quality seeds for planting; observing seed germination; establishing nurseries for crop production.
Summary	Crop propagation is the process of producing new plants from existing plants. The two main types are sexual and asexual propagation. Sexual propagation involves the use of seeds formed through pollination and fertilization. It is widely used because it is simple, economical, and suitable

	for producing large numbers of plants.
Student Engagement **i. What I Have Learnt:** **ii. Skills Acquired:**	
My Question(s):	
Action	

WEEK 9

CROP PROPAGATION

Meaning of Crop Propagation

Crop propagation is the process of producing new plants from existing plants.

It is the method by which farmers multiply crops to ensure continuous production of food, fibre, medicines, and raw materials.

Propagation helps maintain and increase the population of useful crops.

Types of Crop Propagation

There are two main types of crop propagation:

1. Sexual Propagation

This is the production of new plants through seeds formed as a result of pollination and fertilization.

Examples of crops propagated sexually include: Maize, Rice, Cowpea, Groundnut, Tomato, Pepper, Okra

2. Asexual Propagation

This is the production of new plants from vegetative parts such as stems, roots, leaves, and buds without the use of seeds.

Examples include:

- Cassava (stem cuttings)

- Yam (tubers)

- Sugarcane (stem cuttings)

- Banana (suckers)

Sexual Propagation

Sexual propagation is the production of new plants through seeds formed after the fusion of male and female reproductive cells.

It is the most common method of propagating many agricultural crops.

The seed produced contains an embryo that develops into a new plant under favourable conditions.

Importance of Sexual Propagation

Sexual propagation is important because it:

1. Produces large numbers of plants.

2. Is simple and economical.

3. Makes transportation and storage of planting materials easier.

4. Encourages genetic variation among crops.

5. Helps in crop improvement and breeding.

6. Produces crops with strong root systems.

7. It is suitable for large-scale farming.

Processes of Sexual Propagation

Sexual propagation involves a series of biological processes.

1. Flower Formation

Flowers are the reproductive organs of flowering plants.

They contain the male and female reproductive parts needed for seed production**.**

Male Part

- Stamen

Female Part

- Pistil (Carpel)

2. Pollination

Pollination is the transfer of pollen grains from the anther to the stigma of a flower.

Types of Pollination

A. Self-Pollination

Pollen is transferred from the anther to the stigma of the same flower or another flower on the same plant.

B. Cross-Pollination

Pollen is transferred from the flower of one plant to the flower of another plant of the same species.

Agents of Pollination

- Wind

- Insects

- Water

- Birds

- Human beings

3. Fertilization

Fertilization is the fusion of the male gamete with the female gamete to form a zygote.

It takes place inside the ovary of the flower.

The zygote later develops into an embryo.

4. Seed Formation

After fertilization:

- The ovule develops into a seed.

- The ovary develops into a fruit.

The seed contains the embryo that will grow into a new plant**.**

5. Seed Dispersal

Seed dispersal is the scattering of seeds away from the parent plant**.**

Agents of Seed Dispersal

- Wind

- Water

- Animals

- Explosive mechanism

Importance

- Prevents overcrowding.

- Reduces competition for nutrients and sunlight**.**

6. Germination

Germination is the process by which a seed develops into a young plant or seedling under favourable conditions.

Conditions Necessary for Germination

1. Water (moisture)

2. Oxygen (air)

3. Suitable temperature

4. Viable seed

5. Sunlight

Advantages of Sexual Propagation

1. Produces many plants at a time.

2. Seeds are easy to store.

3. Seeds can be transported easily.

4. Encourages crop improvement through variation.

5. Plants often develop stronger root systems.

Disadvantages of Sexual Propagation

1. Some seeds take a long time to germinate.

2. Offspring may differ from the parent plant.

3. Some crops produce seeds with low viability.

4. Plants may take longer to mature.

Key Points to Remember

1. Crop propagation is the production of new plants.

2. The two main types are sexual and asexual propagation.

3. Sexual propagation involves the use of seeds.

4. Pollination and fertilization are important stages in sexual propagation.

5. Fertilization leads to seed formation.

6. Germination produces a new seedling.

7. Sexual propagation is widely used in crop production.

WEEK 10

Topic	**Asexual Propagation**
Objectives / Learning Outcomes	By the end of the lesson, students should be able to: i. Define asexual propagation. ii. Differentiate between natural and artificial asexual propagation. iii. Identify the various methods of natural asexual propagation. iv. Explain the methods of artificial asexual propagation. v. State the advantages and disadvantages of asexual propagation.
Insight	Farmers do not always need seeds to produce new plants. Many crops can be propagated using stems, roots, leaves, buds, and other vegetative parts. Asexual propagation allows farmers to reproduce plants with desirable characteristics and obtain uniform crops within a shorter period.
Key Words / New Terms	Asexual Propagation, Vegetative Propagation, Sucker, Rhizome, Tuber, Bulb, Corm, Cutting, Layering, Grafting, Budding

Application	Propagating cassava using stem cuttings; planting yam tubers; identifying suckers in banana and plantain; establishing SFCSS garden using vegetative propagation methods.
Summary	Asexual propagation is the production of new plants from vegetative parts without the use of seeds. It may occur naturally through structures such as tubers, bulbs, rhizomes, corms, and suckers, or artificially through methods such as cuttings, layering, budding, and grafting.
Student Engagement **i. What I Have Learnt:** **ii. Skills Acquired:**	
My Question(s):	
Action	

WEEK 10
Topic: Asexual Propagation

Meaning of Asexual Propagation
Asexual propagation, also known as vegetative propagation, is the production of new plants from the vegetative parts of a parent plant without the use of seeds.

The vegetative parts used include:

- Stem
- Root
- Leaf
- Bud

The new plants produced are usually identical to the parent plant because there is no fusion of male and female reproductive cells.

Natural Asexual Propagation

Natural asexual propagation occurs naturally without human assistance.

Plants produce special structures that develop into new plants.

The major methods include**:**

1. Propagation by Tubers

Tubers are swollen underground stems or roots that store food.

They possess buds known as "eyes" which develop into new plants.

Examples

- Yam
- Irish potato

Importance

- Food storage
- Plant reproduction

2. Propagation by Bulbs

Bulbs are underground storage organs made up of fleshy leaves surrounding a short stem.

Examples

- Onion
- Garlic

Characteristics

- Store food for plant growth
- Produce new plants from buds

3. Propagation by Rhizomes

Rhizomes are underground horizontal stems that produce roots and shoots.

Examples

- Ginger
- Turmeric

Characteristics

- Grow horizontally beneath the soil
- Produce new plants from buds

4. Propagation by Corms

Corms are swollen underground stems that store food.

Examples

- Cocoyam
- Gladiolus

Characteristics

- Short and solid stem
- Produces new shoots

5. Propagation by Suckers

Suckers are young shoots that grow from the base of the parent plant.

Examples

- Banana
- Plantain
- Pineapple

Characteristics

- Develop from underground buds
- Can be detached and planted separately

6. Propagation by Runners

Runners are slender stems that grow horizontally above the ground and produce new plants at intervals**.**

Examples

- Strawberry
- Lawn grass

Characteristics

- Spread rapidly
- Produce several new plants

Artificial Asexual Propagation

Artificial asexual propagation involves methods carried out by farmers to multiply crops.

These methods are used to improve crop quality and increase production**.**

1. Propagation by Cuttings

Cuttings involve planting a portion of a stem, root, or leaf to develop into a new plant.

Examples

- Cassava (stem cutting)
- Sugarcane (stem cutting)
- Rose (stem cutting)

Advantages

- Easy to carry out
- Produces identical plants

2. Propagation by Layering

Layering involves bending a branch to the ground and covering part of it with soil until roots develop.

The rooted branch is later separated from the parent plant.

Examples

- Bougainvillea
- Jasmine

3. Propagation by Budding

Budding involves inserting a bud from one plant into another closely related plant.

Examples

- Citrus crops
- Roses

Importance

- Improves crop quality
- Enhances disease resistance

4. Propagation by Grafting

Grafting involves joining the stem of one plant (scion) to the root system of another plant (rootstock) so that they grow as one plant.

Examples

- Mango
- Orange
- Avocado

Importance

- Combines desirable characteristics
- Produces improved crops

Advantages of Asexual Propagation

1. Produces plants identical to the parent.
2. Maintains desirable characteristics.
3. Plants mature and produce earlier.
4. Useful for crops that produce few or no viable seeds.
5. Produces uniform crops.
6. Enables rapid multiplication of plants**.**

Disadvantages of Asexual Propagation

1. Little or no genetic variation.
2. Diseases can be transferred from parent plants.
3. Some methods require special skills.
4. Planting materials may be bulky to transport.

5. New varieties are not easily developed**.**

Differences Between Sexual and Asexual Propagation

Sexual Propagation	Asexual Propagation
Uses seeds	Uses vegetative parts
Involves fertilization	Does not involve fertilization
Produces variation	Produces identical offspring
Common in maize and rice	Common in cassava and yam
Usually slower to maturity	Usually faster to maturity

Key Points to Remember

1. **Asexual propagation is the production of new plants without seeds.**
2. **It uses vegetative parts such as stems, roots, leaves, and buds.**
3. **Natural methods include tubers, bulbs, rhizomes, corms, suckers, and runners.**
4. **Artificial methods include cuttings, layering, budding, and grafting.**
5. **Asexual propagation produces plants that are genetically similar to the parent plant.**
6. **It is widely used for crops such as cassava, banana, plantain, sugarcane, and many fruit trees.**